

orbits - practice problems

name _____

1. A 40 kg rider on a carnival ride moves in a circle with a radius of 8.0 m at 6.0 m/s. What centripetal force is required, and which direction does it point?
2. A car drives around a flat, circular highway curve with a radius of 75 m. The coefficient of static friction between the tires and the dry road is 0.80. What is the fastest the car can go around the curve without sliding?
3. A rotating space station design has radius 40 m, about the length of a large classroom hallway. How fast does the outer edge need to move to create an artificial gravity acceleration of 9.8 m/s^2 ?
4. The outer edge of the 40 m radius station moves at 19.8 m/s. How long does one rotation take?

5. Spinning faster than about 2 rotations per minute makes many people dizzy. A space station is designed to rotate once every 30 s and produce 9.8 m/s^2 at its outer edge. What radius does it need?

6. A ball on a string is swung in a horizontal circle. At one moment the string breaks. Which way does the ball go?

7. A small satellite is 100 km above the Moon's surface. Use the Moon's mass as $7.346 \times 10^{22} \text{ kg}$ and radius as $1.737 \times 10^6 \text{ m}$. What circular orbital speed is needed?

8. Astronomers discover a planet with a radius of $5.0 \times 10^6 \text{ m}$. A 500 kg probe is put into a circular orbit 300 km above its surface. What is the probe's orbital speed?

9. A satellite is 400 km above Earth's surface. Use Earth's mass as $5.972 \times 10^{24} \text{ kg}$ and radius as $6.371 \times 10^6 \text{ m}$. What circular orbital speed is needed?

10. A very low circular orbit around Mars has radius 3.390×10^6 m from Mars's center. Use Mars's mass as 6.417×10^{23} kg. What orbital speed is needed?

11. A 100 kg satellite and a 400 kg satellite both orbit Earth at a radius of 1.00×10^7 m. First find the orbital speed, then find the centripetal force needed for each satellite. Use Earth's mass as 5.972×10^{24} kg.

12. A satellite around Earth has a circular orbital speed of 6000 m/s. Use Earth's mass as 5.972×10^{24} kg. What is the orbital radius, and what is the altitude above Earth's surface?

13. A GPS satellite has an orbital speed of about 3870 m/s. Use Earth's mass as 5.972×10^{24} kg and Earth's radius as 6.371×10^6 m. Estimate its orbital radius and altitude.

14. Use the 400 km altitude Earth satellite from earlier. Its orbital radius is 6.771×10^6 m and its speed is 7670 m/s. How long does one orbit take?

15. Around the same planet, satellite B orbits at 4 times the orbital radius of satellite A. How does satellite B's circular orbital speed compare to satellite A's?

16. A satellite around Earth has orbital radius 4.22×10^7 m and speed 3070 m/s. If another circular orbit around Earth has one-fourth that radius, what speed would it need?

17. A 900 kg satellite is in a 400 km altitude circular orbit around Earth. Its orbital radius is 6.771×10^6 m and its speed is 7670 m/s. What centripetal force is required?

18. A satellite is at an orbital radius of 1.00×10^7 m from Earth's center and is moving at 5500 m/s. Use Earth's mass as 5.972×10^{24} kg. Is that fast enough for a circular orbit at that radius?

19. A satellite orbits 800 km above the Moon's surface while mapping crater shadows. Its orbital radius is 2.537×10^6 m and its circular speed is 1390 m/s. Estimate the time for one orbit.